sanitary inspectors and laboratory technicians) who work on disease prevention and health promotion must be recognised and appreciated.

Many countries have conducted reviews of their pandemic responses to identify gaps—these were mainly in the area of public health—and to ensure early detection, quality-assured laboratory infrastructure for diagnosis of known and new pathogens, trained manpower, real-time data collection and analysis, a risk communication strategy and empowered local and national health authorities who could advise the administration on action needed. Paradoxically, we observed that some low- and middle-income countries, which relied less on hi-tech and more on existing primary health and community-based approaches (like Rwanda, Senegal, Thailand and Vietnam) did relatively well. Investments in primary healthcare, including regular training and mentoring of the workforce and the use of digital tools to support them, are going to be important. India's attention to digital infrastructure, with the National Digital Health Blueprint defining the standards as well as governance and vision, put it in a better position as it was quickly able to roll out mobile applications for contact tracing, vaccination and telemedicine services. These digital health services should be expanded and strengthened to leverage the presence of excellent specialists in our cities who can also serve remote rural populations and thereby improve the quality of healthcare.

We now know much more about the SARS-CoV-2 virus and its modes of transmission, so our actions must be strategic in future. Restrictions on travel (and lockdowns) have minimal benefits and should be imposed only as a temporary measure. Attention to ventilation is important for all airborne respiratory pathogens. Either good natural ventilation or upgraded HVAC systems should be in place in all settings where a lot of people spend time indoors, such as educational institutions, workplaces, malls and restaurants. We need to employ technology much more smartly. Genomic surveillance is one example. Many countries made use of it to understand their epidemics—where the virus entered, when it entered, how many points of entry it had in a country, how it spread and how it is evolving. The Indian government set up INSACOG, a consortium of institutions, for genomic surveillance. It has been tracking the virus and its variants across India. Thought should now be given as to how to use this system to tackle other infectious diseases.

THE PANDEMIC SHOWED US **RESTRICTIONS ON TRAVEL HAVE MINIMAL BENEFITS.** ATTENTION TO VENTILATION IS IMPORTANT. TECHNOLOGY MUST BE USED SMARTLY

The unprecedented development of several safe and effective vaccines has enabled the world to exit the acute phase of the pandemic. It was previously unimaginable that not only would vaccines be developed and deployed within a year of the emergence of a new virus, but that about 14 billion doses would be administered in two years globally. Both R&D and manufacturing scaling surpassed all expectations. Covax set up by WHO with CEPI, Gavi and UNICEF played a key role in addressing inequity, with nearly 2 billion doses deployed by the end of 2022, mostly to low- and lower middle-income countries. The Indian vaccine industry played a key role in providing vaccines for the huge Indian population as well as manufacturing for the world. For the future, attention needs to be paid to how the intellectual property of lifesaving products like vaccines and drugs is managed during a global emergency—rapid technology transfer and a quality-assured network of distributed manufacturing sites (including on the African continent) is the only way to meet the demands of the whole world quickly. It is estimated that over a million lives could have been saved in 2021 alone if vaccine access had been more equitable. The pandemic accord being negotiated by WHO member-states is a once-in-a-lifetime opportunity to address this issue, so that the same mistakes are not repeated.

The pandemic has taught us many lessons and we need to learn them to be better prepared next time. The question over another pandemic is not if, but when, it will occur. The world is facing many challenges, including the impacts of climate change on health, food systems and biodiversity, related environmental risks like air pollution and water scarcity, as well as the increasing threat of anti-microbial resistance. We need to take a holistic view of health, which encompasses the One Health dimension as well as focuses on the prevention of disease and health promotion (both physical and mental). If the global scientific and public health community work together and agree to share knowledge, data and solutions, these challenges can be overcome and we can get back on track to achieve the Sustainable Development Goals. One motto, we must not forget: Leave No One Behind. ■